

Logistic Regression

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- The statisticians approached the problem - “**how can we use linear regression to solve this?**”
 - We could consider the following encoding
 - Dependent variable coded as 0 (**Not Spam**) or 1 (**Spam**)

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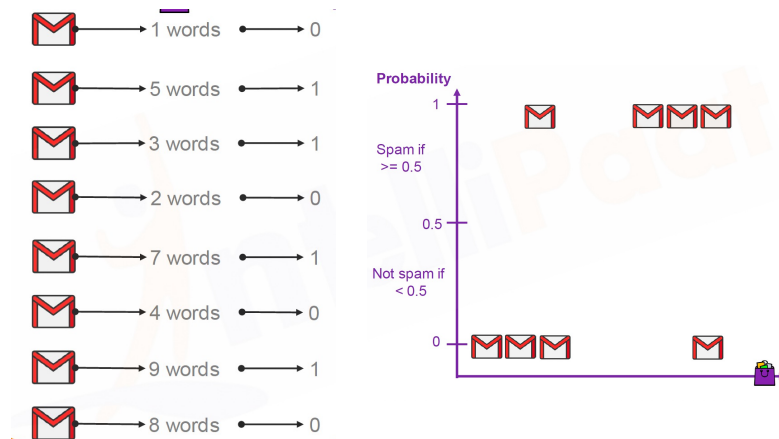
- We can fit a linear regression to this binary response
 - and classify as **Spam** if $\hat{y} > 0.5$ and **Not Spam** otherwise, interpreting $\hat{y}$ as a probability that **Email is Spam**

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• Spam email classifier



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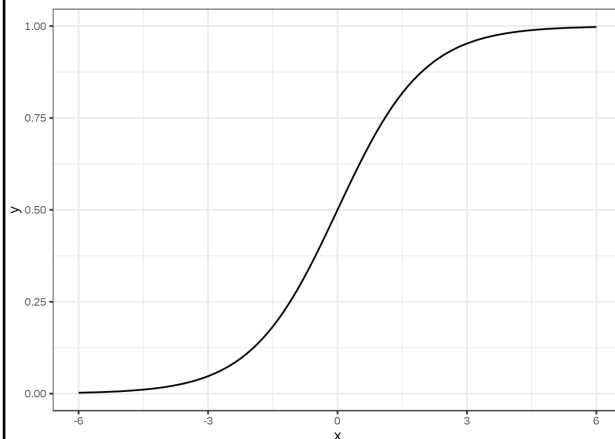
- A major problem with such an approach
- Linear regression models produce values in $(-\infty, +\infty)$, which does not make sense as a probability
 - Employ a function that constrains the values between 0 and 1

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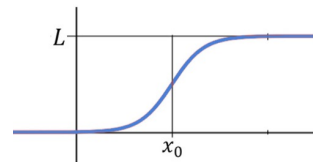
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- Logistic function (Sigmoid)



$$f(x) = \frac{L}{1 + e^{-k(x-x_0)}}$$

x_0 = x value of midpoint
 L = maximum value
 k = growth rate



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- Logistic regression model

Sigmoid output $\rightarrow$ $p_X = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} \quad 0 \leq p_x \leq 1$ $\leftarrow$ Linear regression output



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- Terminology

- The **logit** $\log\left(\frac{p_X}{1-p_X}\right)$ ← Log **odds**

- The odds of an event is defined as

$$\text{odds}(Y = 1) = \frac{P(Y = 1)}{1 - P(Y = 1)} = \frac{p}{1 - p}$$

- Chance can be expressed either as a probability or as odds

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- The cost/loss function (compares the model's **predictions** to the **target values**)

- The **likelihood function**

$$(1 - P_{x1}) \times (P_{x2}) \times (P_{x3}) \times (1 - P_{x4}) \times \dots$$

- The likelihood is a function of model parameters (β_0, β_1) and we can estimate them by **maximizing the likelihood**

- Maximum likelihood estimation (MLE)

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- The cost/loss function
 - No closed form solution for MLE
 - We rely on numerical approximation to find the MLE
 - Most software uses the [gradient descent algorithm](#)

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- In practice, it is more convenient to maximize the [log of the likelihood function](#)
 - product of a large number of small probabilities can easily lead to [underflow in computing machines](#)

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